Hybrid Retrieval for Short Scientific Questions: A Controlled Benchmark

A. Example and B. Sample | Demonstration Research Group | 2024 (fictional example)

Abstract

This fictional study compares a hybrid retrieval pipeline with a lexical baseline for short questions over research documents. The benchmark contains 1,200 synthetic question-document pairs from four subject areas, with a fixed 960-pair development set and 240-pair test set. The hybrid system combines BM25 candidate retrieval with a compact dense reranker. On the test set, it reaches NDCG@10 of 0.742 and Recall@10 of 0.812, compared with 0.681 and 0.754 for BM25 alone. Median latency rises from 92 ms to 185 ms per query. These results suggest a ranking improvement within this benchmark, with a latency cost. The experiment uses synthetic data, one split, and no external replication; the measurements do not establish performance on real literature collections.

1. Introduction

Searching research papers often begins with a short question whose wording differs from the terms used in a relevant passage. Lexical retrieval is efficient and interpretable, but exact term overlap can miss paraphrases. This demonstration evaluates whether a dense reranking stage improves a lexical candidate list while keeping the candidate-generation step unchanged.

Research question: How do the tested retrieval methods compare on ranking quality and latency in this fixed synthetic benchmark?

2. Methods

The lexical baseline uses BM25 with k1 = 1.2 and b = 0.75. It returns the top 100 passages. The hybrid system applies the same BM25 stage, then reranks those candidates with a compact dual-encoder similarity model. Both systems use identical document text, tokenization, test questions, and relevance labels. No query expansion or external source is used.

3. Dataset and evaluation

The synthetic benchmark contains 1,200 question-document pairs across computing, biology, materials science, and public health. Each question has between one and five relevant passages. We assigned 960 pairs to development and 240 to test before tuning. Development data was used for parameter selection; the test set was evaluated once. All scores below are illustrative values created for application testing.

We report NDCG@10 and Recall@10 for ranking quality, plus median per-query latency. Both ranking metrics range from 0 to 1, with higher values indicating better performance. Latency is measured in milliseconds, with lower values indicating faster responses. No uncertainty interval is available.

4. Results

System	NDCG@10	Recall@10	Median latency
BM25 baseline	0.681	0.754	92 ms
BM25 + dense reranker	0.742	0.812	185 ms

The hybrid system improves NDCG@10 by 0.061 and Recall@10 by 0.058 relative to BM25 in this test split. It also has approximately twice the median latency. The evaluation does not isolate which subject area contributes most to the change, and it does not report confidence intervals or statistical significance tests. The point estimates should therefore be read as descriptive benchmark results.

5. Limitations

The collection and labels are synthetic and may not reflect real user questions, scholarly writing, or relevance judgments. Results come from a single fixed split and a single compact reranker. We did not measure hardware variation, long-document behavior, multilingual retrieval, or downstream answer quality. No claim is made that this system is generally superior.

6. Conclusion

Within this fictional benchmark, dense reranking improves the reported ranking metrics and increases latency. A real deployment decision would require evaluation on representative documents and queries, repeated splits, uncertainty estimates, and a target-specific latency budget.

Data and provenance note

All study names, authors, data, metrics, and findings in this PDF are fictional and generated solely for application testing. They must not be cited as real research.